vCRGlove Project Handbook

A readable companion to the project documentation

Setup, data formats, release steps and physical testing. This PDF is generated from the Markdown sources so the two formats can stay consistent.

Review date 2026-10-21. This is an implementation reference, not patient information or a clinical protocol.

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vCRGlove

iPhone and Apple Watch research app for **vibrotactile Coordinated Reset (vCR) stimulation**, hand-movement recordings, and Parkinson's symptom journaling.

Developed at the **Institute for Computational Neuroscience (ICNS), University Medical Center Hamburg-Eppendorf (UKE)**.

Status: active research development. Movement measurements are not validated clinical ratings, and the app does not assign MDS-UPDRS scores.

Current Features

Stimulation

- Bluetooth scanning and left/right connection management for **bHaptics TactGlove DK2**.
- Patient session controls, including pause/resume and hold-to-stop, with connection and interruption feedback.
- Continued stimulation with one connected glove; automatic interruption when both disconnect.
- One vCR tab with saved per-hand amplitudes, cycle frequency and pattern controls after researcher unlock; staff Settings contains the logs.

Movement Tasks

- Finger tapping and hand opening/closing for each hand, captured with the iPhone camera.
- Pronation/supination using live motion data from the Apple Watch companion.
- Guided hand-size calibration, silent demonstration videos for camera tasks, preparation countdown, and live tracking feedback.
- Timed 30-second research recordings, with early stopping and retake options.
- Automatic local saving of completed recordings in the guided session flow, without requiring every task to be finished.
- Recording timestamps, stimulation context, and patient-reported medication timing.

Results and Data

- Daily-median trends for rate, camera movement size and rhythm variability, separated by task, hand, context and measurement method.
- Tap a day for counts/ranges and its recordings; pauses and amplitude change are under More details.
- Date, stimulation-context, and medication-context filters, plus a separate recording history.
- Measurement export and private-video playback, sharing, and deletion, including multi-video selection.
- Researcher-only full-data ZIP sharing: journal, stimulation logs, raw movement records, saved videos and study settings, with a checksum manifest.
- Unavailable measurements are shown as unavailable rather than plotted as zero.

Journal and Settings

- Calendar with an entry legend and a daily timeline.
- Quick check-in for mood and the effect of Parkinson's on usual activities.
- Direct medication, symptom-episode and note logs, with editing and local persistence.
- English/German interface using one string catalog, with an in-app language selector.
- Text-size settings, instructions, and troubleshooting guidance.
- Bilingual study information and data-protection notice, plus factual private-video storage information.

Getting Started

Deployment targets are **iOS 17+** and **watchOS 10+**. Use an Xcode version that supports the operating systems installed on your test devices.

1. Clone the repository and open `vCRGlove.xcodeproj` in Xcode.
2. Select the `vCRGlove` scheme and configure Signing & Capabilities for your authorized development team.
3. Build and run on a physical iPhone for camera and Bluetooth testing.
4. Turn on the TactGlove DK2 gloves and scan for them inside the app.
5. Install the Watch companion for pronation/supination testing. The Watch is not required for iPhone camera tasks or glove control.

The bHaptics SDK is included as a local Swift package at `Vendor/bhaptics-ios-0.2.0`. Bundled demonstration videos are in `vCRGlove/Resources/hand_videos`; no separate download is needed.

Simulator tests can check interface and software behaviour, but do not replace physical camera, glove, and Watch testing.

Storage and Privacy

Data is stored inside the app's container on the iPhone:

Data	Location
Stimulation and app events	`Documents/vcr/logs/events.jsonl`
Journal entries	`Documents/vcr/journal/journal_entries.json`
Movement sessions and signals	`Documents/vcr/tasks/sessions.jsonl`
Recorded camera-task videos	`Application Support/vcr/tasks/videos/`

Camera-task videos are silent and are not automatically saved to Photos. Video storage requests iOS file protection and is excluded from automatic backup. **Local storage is not a backup:** deleting the app or losing the device can lose recordings.

Use **Settings > Data** to review, share, or delete recorded videos. Deleting a video keeps its movement measurements and does not delete copies previously shared elsewhere. JSON/CSV measurement

exports do not contain video files. Researchers can unlock **Study team > Study setup & logs > Share all recorded data** to share one ZIP including all retained task movies and study records. Sharing is manual, contains sensitive data and does not verify recipient receipt or backup.

Backend upload code exists but is not configured at app launch. It is not yet a completed consent-based research-sharing workflow, and that upload path does not transfer video files. Settings includes study information and the adapted privacy notice; institutional review and additional patient controls remain pending. See source and consent boundaries.

Next Priorities

- Review study/privacy wording and complete consent controls, including a choice not to store videos.
- Validate camera measurements, new velocity metrics and task-specific composite scores against recordings and clinical ratings.
- Add configurable journal reflection days/reminders and streak feedback. PDQ-8 remains planned, subject to permissions and study requirements.
- Verify recipient-side backup and recovery and agree consent-based transfers; the manual researcher ZIP is not an automatic upload or verified backup service.
- Complete long-duration stimulation, reconnect, storage, and physical Watch validation.

Documentation

Start with the documentation index for setup, code ownership, data formats, feature references and known limitations.

- Release and Git checklist
- Physical-device testing checklist

Research Background

vCR research investigates whether patterned fingertip stimulation can support desynchronization of abnormal neuronal activity. Relevant publications include:

- Popovych, O. V., & Tass, P. A. (2012). Desynchronizing electrical and sensory coordinated reset neuromodulation. *Frontiers in Human Neuroscience*, 6:58.
- Pfeifer, K. J., et al. (2021). Coordinated Reset Vibrotactile Stimulation Induces Sustained Cumulative Benefits in Parkinson's Disease. *Frontiers in Physiology*, 12:624317.

These publications inform the research direction; they do not validate this app, its glove implementation, or its movement scores.

License

This project is part of ongoing research at ICNS, UKE. Licensing terms are under review; **all rights reserved** for now. Please contact the authors for research collaboration inquiries.

Project Documentation

Current implementation reference, reviewed 16 October 2026. Start here for development and team handover. Patient guidance lives inside the app.

Prefer a document reader? Open the PDF handbook. It contains the overview, setup, data formats, release and testing chapters, with a clickable contents page. Markdown remains the editable source.

Start Here

Need	Reference
Understand the app	Project overview
Set up Xcode, find the relevant code, run tests	Development guide
Understand saved records, timestamps and exports	Data formats
Prepare a reproducible TestFlight build	Release checklist
Test on an iPhone, gloves and Watch	Physical testing checklist

Feature References

Area	Reference	Status
Movement instructions and videos	Demonstrations	Implemented; hardware checks remain important
Researcher stimulation settings	Shared vCR controls	Implemented; per-hand setup and timing require physical validation
Manual full-device data sharing	ZIP contents and limits	Researcher-only manual export; verified backup/consent-based transfer remain separate
Daily trends and filters	Results	Implemented; camera/clinical validation remains pending
Private recordings and deletion	Video storage	Implemented; verified backup and retention remain pending
Task-specific composite measures	Research review	Proposal, not a validated score
Auxiliary tapping angles	Stages and validation	Capture/storage/export implemented; physical and external-reference validation pending

Area	Reference	Status
In-app tapping angle measurements	Definitions and compatibility	Additive angle summary; existing trends unchanged; healthy-reference validation pending
Offline angle comparison	Run and interpret the review	Six-trial paired-channel review complete; broader accuracy validation pending
Author-derived tapping features	ROC-HCI feature subset	Separate offline kernel verified against pinned code; no patient bands
Reference-pipeline bridge	Paired landmarks and first ten taps	Comparison-window metadata implemented; research validation is not a healthy reference band
Independent tapping-video review	Findings and review tools	Initial stress-test review complete; broader camera accuracy/orientation validation remains
Camera movie/signal alignment	Timing metadata	Anchor implemented; initial two-clip phone check passed; broader repeatability/lifecycle checks pending
Healthy movement references	Dataset and protocol assessment	Research review complete; external controls not yet validated for patient bands
Journal redesign	Stages and verification	Stages 1/2 implemented; device review pending
Journal design rationale	Original proposal	Design history, not the current screen specification
Medication/activity integration	Apple Health review	Decision pending; no integration added
Study information in Settings	Content and source	Bilingual information pages; legal review/patient controls remain separate
Backend interface	Integration reference	Dormant client path; incomplete server fields/consent workflow

Working Backlog

The shared planning file is `QA/app_improvement_backlog.xlsx`, maintained separately from Git. Use its stable IDs, such as `MOV-005` or `JRN-002`, when discussing work. A missing workbook in a fresh clone is expected; obtain the current copy from the project team.

Keep existing formatting and manual statuses when updating it. Close Excel before file-based edits. Record what was implemented, what was verified, and what still needs a device check or approval. Do not mark an entire staged redesign complete after finishing only one stage.

Historical Material

These records explain earlier decisions; their commands, test counts and feature descriptions may be outdated:

- Original movement-module handoff.

- July analyzer test report.

- July camera changelog.

- July Watch changelog.

- Backend prototype and clinic prototype. These are not a deployed patient-data service.

Synthetic examples and tests are software fixtures, not clinical reference populations or validation of a score.

Keeping This Useful

- Update the relevant feature document and checklist with the code change. Link from this index instead of copying the same explanation into several files.

- For storage/schema changes, update Data formats and add compatibility/failure tests before migrating data.

- Record release-specific evidence in the release record. Passing tests for an older commit are not evidence for a later build.

- Keep proposals clearly separate from implemented behaviour. Code and reproducible tests resolve implementation discrepancies; approved study documents govern clinical and privacy requirements.

- Keep patient data, identity mappings, consent forms, credentials and signing keys out of the repository. Store sensitive operational documents in approved institutional storage.

Rebuild the PDF

After a major change, update the affected source chapters and regenerate the PDF. Use the bundled workspace Node/Python runtimes, with Marked, ReportLab and pypdf available. The builder resolves those dependencies beside the bundled Node executable; `CODEX_NODE_MODULES` and `CODEX_PYTHON` can override their locations.

Run from the repository root, replacing the example date with the actual review date:

```
node tools/build_documentation.mjs --date 2026-10-15
node tools/build_documentation.mjs --date 2026-10-15 --check
```

The check compares the PDF's source fingerprint with all included Markdown chapters. It detects stale source content, not layout or correctness. Render and inspect the rebuilt pages before delivery. The PDF contains no ethics source documents or patient records.

Development Guide

Implementation reviewed 15 October 2026. See the documentation index for feature details and pending decisions.

Setup and Signing

1. Open `vCRGlove.xcodeproj`. The local bHaptics package is included in `Vendor/bhaptics-ios-0.2.0`; demonstration assets are bundled in `vCRGlove/Resources/hand_videos`.
2. Select the `vCRGlove` scheme. The deployment targets are iOS 17 and watchOS 10. Check Apple's Xcode compatibility matrix for the Mac, Xcode and device combination. Deployment minimum, bundled SDK and physical-device support are separate constraints.
3. Use the authorized ICNS team with automatic signing. Current team ID: `X62VCFT697`. Phone bundle ID: `de.uke.icns.vcrglove`; companion: `de.uke.icns.vcrglove.watchkitapp`. Keep these identifiers when updating the existing app. Do not create new IDs to resolve a launch error.
4. Check each app and test target's Signing & Capabilities. Enable only capabilities required by its code/entitlements. Camera/Bluetooth privacy descriptions and background modes are not a reason to select unrelated Developer Portal services.
5. Run the iPhone app on a physical phone for camera and glove checks. Watch development can be tested separately on a compatible Mac/device pair. A successful iPhone run does not establish Watch debugging support.

Do not commit certificates, profiles, private keys or personal Xcode workspace settings. Current identity values are configuration, not secrets. Access and signing authority still belong to the development team.

Code Map

Paths below are relative to the repository root.

Responsibility	Main files
Launch, language and tab order	`vCRGlove/vCRGloveApp.swift`, `vCRGlove/MainTabView.swift`
Glove control and patient sessions	`vCRGlove/GloveVM.swift`, `vCRGlove/PatientVCRView.swift`
Researcher controls and stimulation profile	`vCRGlove/ResearchTools.swift`, `vCRGlove/StimulationProfile.swift`
Movement flow, demo and capture	`vCRGlove/MovementTaskView.swift`, `vCRGlove/MovementDemonstrationView.swift`, `vCRGlove/VisionHandPoseCapture.swift`
Samples, calculations and models	`vCRGlove/TrialRecorder.swift`, `vCRGlove/MovementAnalyzer.swift`, `vCRGlove/MovementModels.swift`

Responsibility	Main files
Movement persistence, video and export	`vCRGlove/TaskSessionStore.swift`, `vCRGlove/TaskSessionExporter.swift`, `vCRGlove/MovementExportView.swift`
Researcher full-data ZIP	`vCRGlove/FullDataExporter.swift`, raw snapshots in journal/task/event stores
Movement results	`vCRGlove/MovementTrendView.swift`
Journal screens and persistence	`vCRGlove/JournalHomeView.swift`, `vCRGlove/DailyCheckInView.swift`, `vCRGlove/JournalStore.swift`, `vCRGlove/JournalModels.swift`
Settings, study information and support	`vCRGlove/SettingsView.swift`, `vCRGlove/Localizable.xcstrings`
Event logging	`vCRGlove/EventStore.swift`, shared `Logger` in `vCRGlove/ResearchTools.swift`
Watch transport and capture	`vCRGlove/PhoneWC.swift`, `vCRGlove/WatchMotionCapture.swift`, `vCRGloveWatch Watch App/MotionService.swift`
Upload prototype	`vCRGlove/SessionUploader.swift`
Tests	`vCRGloveTests/`, `vCRGloveUITests/`

Tab order is always vCR, Journal, Movement, Settings. Researcher unlock never adds a tab; the old `showResearchTab` preference is ignored. English/German strings belong in `vCRGlove/Localizable.xcstrings`; `L10n` looks them up using the in-app language choice. Do not restore a separate translation dictionary.

Language refresh retains the selected tab and General route in app-owned state above `.id(appSettings.language)`. It stays in Settings > General; relaunch starts normally. Profile shows the study ID only; avatar customization is deferred.

Research study-ID edits are staged until Save. Changing an assigned ID requires confirmation; an assigned ID cannot be cleared. Cancel/back leaves it unchanged. Only future recordings use a confirmed new ID; historical records are never relabelled or automatically merged. Optional usernames remain deferred and must not become research identity keys. Profile's ID remains read-only.

For a TEAM ID, unlocked Research Admin offers an explicit, reversible display link for the researcher's own earlier unassigned tests. Confirming snapshots existing session/trial UUIDs only; it never automatically links future tests or assigned identities. `movementTrendIdentityLinks` stores the local JSON mapping in app preferences. Results observes it, while session stores and JSON/CSV exports retain original IDs. The full research archive includes the mapping in `study-settings.plist` for provenance. Do not bundle real session IDs or enable automatic `unset` aliases for participants.

Behaviour to Preserve

- Patient stimulation offers 4 hours or two 2-hour sessions. One disconnected glove does not end stimulation on the other; losing both interrupts the session. Verify actual hardware behaviour, not just the displayed state.

- The idle timer is disabled during active stimulation. Inactive scene transitions are logged; background transitions also show a timing warning and request a finite background-task grace period. This does not guarantee foreground priority or uninterrupted stimulation when locked/backgrounded.

- Camera tasks use a fixed 30-second research interval, with early Stop. Missing hand tracking leaves gaps in the signal; it does not pause the recording clock. Capture feedback and analysable movement are different checks.

- A completed guided take is checkpointed locally before continuing. Retake preserves the old take until the replacement is saved. An unfinished draft can still be discarded; exact navigation restoration after process termination is not implemented.

- Stimulation context is selected for the session; medication timing is confirmed at the start and attached to each trial. Missing/unsure context must not become a negative answer or a medication ON/OFF classification.

- Journal check-in now asks mood and activity impact. Medication, symptom episodes and notes have separate log entry points and editing. Existing answers/unknown fields are preserved; reminder schedules and PDQ-8 are not implemented by this redesign stage.

- Private movies are silent and are not automatically placed in Photos. Review save and export limits before changing storage or sharing.

- The primary Settings order is Profile, General, Instructions, vCR Settings, Reminders, Data, Information about the study. General groups the existing language and text-size preferences; Profile no longer contains text size and the separate globe shortcut is removed. Text-size choices use full-width rows for readable large-text labels. Existing saved preferences are unchanged. Info is read-only; Contact is its last topic and contains contact details only. Devices and app contains the factual storage/video explanation formerly under Privacy & Data. The legal notice remains separate and unchanged. Reading these pages is not consent. See source and review boundaries.

Shared vCR and Researcher Controls

Settings ends with a discreet Study team entry below Contact/Help. Unlock exposes Study setup & logs in Settings and a collapsible Stimulation parameters panel on the existing vCR page. The patient view has no parameter sliders or logs when locked. `VCRView.swift` was removed; its shared logger and log panel now live in `ResearchTools.swift`.

Researchers set left/right SDK amplitude (0-100), cycle frequency (0.5-2.5 Hz) and either shuffled single-finger or legacy all-fingers-together stimulation. Both retain four slots per cycle and 100 ms pulses. Shuffled uses SDK motor indices 0-3 once each in random order; simultaneous drives the existing 20-element motor array on every slot, not just once per cycle. The upper frequency limit prevents scheduled 100 ms pulses overlapping; it is a software constraint, not clinical validation or a recommended treatment range.

Save is explicit and is rejected while running, paused or checking fingers. The saved profile survives restart and is used by patient Start, the five-motor startup check, idle test buzz, resume and a joining glove. Fresh installations retain the previous 70/70 amplitude, 1.5 Hz shuffled preset until staff save another profile. Staff must set and verify each patient's amplitudes in person before handover; there is

no new mandatory setup gate or automatic calibration.

Each session snapshots the profile. Locking controls does not reset it or change stimulation. The app owns the controller above the language-dependent view identity, so changing language does not replace it. Stop/Pause invalidate scheduled slots; stale pulse-off callbacks cannot switch off a newer pulse. Stop during the startup check cancels its asynchronous continuation. A glove disconnection during the startup check stops the patient session and asks for a fresh Start rather than silently skipping the check. A corrupt/unsupported stored profile blocks stimulation until staff review and save it; defaults are not silently substituted for delivery.

Staff Settings retains patient ID/study metadata, the visible log, verbose BLE diagnostics and event-log sharing/clearing. Share all recorded data now prepares one ZIP containing raw journal, stimulation/app events, task signals/metrics, saved movies and study settings, with a checksum manifest. It opens the native share sheet without automatic upload or source deletion. Export is disabled during active/paused/startup stimulation. This is a manual export, not a verified backup or consent-based research transfer; see Data formats.

Unlock is in memory and starts locked on relaunch. The existing hard-coded password remains a convenience UI gate, not secure authentication or proof of researcher identity. Do not describe it as a security boundary. Saved profiles are local configuration, not a signed prescription or a complete study audit trail.

Run Tests

In Terminal, from the repository root, first inspect the available simulators:

```
DEVELOPER_DIR=/Applications/Xcode.app/Contents/Developer xcrun simctl list devices available
```

Choose an installed iPhone simulator and replace `SIMULATOR_UDID` below. `DEVELOPER_DIR` selects the full Xcode app for this command only; adjust its path if Xcode is installed elsewhere.

```
DEVELOPER_DIR=/Applications/Xcode.app/Contents/Developer xcodebuild test \
  -project vCRGlove.xcodeproj -scheme vCRGloveTests \
  -destination 'platform=iOS Simulator,id=SIMULATOR_UDID' \
  -derivedDataPath /tmp/vcr-unit-derived
```

For the focused Journal UI flows:

```
DEVELOPER_DIR=/Applications/Xcode.app/Contents/Developer xcodebuild test \
  -project vCRGlove.xcodeproj -scheme vCRGloveUITests \
  -destination 'platform=iOS Simulator,id=SIMULATOR_UDID' \
  -only-testing:vCRGloveUITests/vCRGloveUITests/testJournalTwoQuestionsAndSavedLogShortcutsEng\
lishGerman \
  -only-testing:vCRGloveUITests/vCRGloveUITests/testJournalEmptyShortcutDoesNotCreateCheckInEn\
glishGerman \
  -only-\
testing:vCRGloveUITests/vCRGloveUITests/testJournalFailedSaveKeepsCheckInAnswersEnglishGerman \
\
  -only-testing:vCRGloveUITests/vCRGloveUITests/testJournalCheckInCanBeEditedAndSurvivesRelaun\
chEnglishGerman \
  -only-\
testing:vCRGloveUITests/vCRGloveUITests/testJournalMedicationAndSymptomEditsEnglishGerman \
  -only-testing:vCRGloveUITests/vCRGloveUITests/testJournalSelectedDayAndFutureDateEnglishGerm\
anLargeText \
  -derivedDataPath /tmp/vcr-ui-derived
```

Use the test target's launch harness and synthetic fixtures, not patient files. Record the commit, toolchain, destination, command and result bundle with each verification. Do not run competing test jobs against the same simulator.

For the bilingual study-information navigation and long-text check, use the UI test selector `testStudyInformationAndSupportEnglishGermanLargeText` in the same UI test class. It visits all six topics, the institutional support address and factual app-video storage text; German uses the largest app text size.

`testGeneralSettingsGroupsLanguageAndTextSizeEnglishGerman` covers placement, font choices, persistence and ID-only Profile. `testLanguageSwitchStaysInGeneralEnglishGermanLargeText` covers repeated switches, selected tab and Back at normal/extra-large sizes.

The focused researcher UI selector is `testResearchControlsShareVCRPersistAndRelockEnglishGerman`. It checks the fixed four tabs, unlock placement/password rejection, per-hand edits, pattern selection, persistence and relocking in English and large German text. `testResearchPatternMenuWrapsAndSelectsEnglishGermanLargeText` checks the wrapping pattern menu and Save; `testLanguageChangeRetainsSharedResearchController` checks controller retention through a language change. `StimulationProfileTests` uses isolated settings and injected motor/event outputs, without physical glove commands.

Full-device export checks use `FullDataExportTests`, `FullDataArchiveTests` and UI selector `testResearchFullDataShareEnglishGerman`. ZIP fixtures contain synthetic data only. Verify extracted file hashes, raw unknown/malformed records, missing-video notices, credential exclusion, empty-device export and temporary-copy cleanup. Do not use patient archives as test attachments.

Validation Limits and Open Work

Simulator tests cover calculations, compatibility, save failures and UI flows. They do not validate camera tracking accuracy, glove timing, long-duration operation, Watch orientation/streaming, physical file protection or clinical usefulness. Use the physical checklist and approved non-patient fixtures before study deployment.

The 14 October Journal work passed 135 unit tests and six distinct focused UI flows across runs, including English/German and large text. Those results belong to that working snapshot, not future releases. Physical review is pending; the existing symptom grid needs particular attention at the largest text size.

The 15 October Settings-refinement snapshot passed 139 unit tests and one focused English/German UI test on the iPhone 17e iOS 26.5 simulator, including menu order, contact-only navigation, relocated data text, separate troubleshooting and the largest German app text size. The retained German privacy copy was separately compared with Word. Results and source boundaries are recorded in Study information; institutional review and physical reading checks remain pending.

The later 15 October shared-vCR snapshot passed 154 Swift Testing tests (plus the existing XCTest rendering test). Three focused UI checks passed across targeted runs: researcher configuration/persistence/relocking, the wrapping English/German pattern menu, and controller retention through a language change. The largest German app text size was included. An unsigned generic-iPhone Release build also succeeded with Xcode 26.5. This does not verify signing, installation, physical amplitudes, motor mapping or pulse timing; those remain bench/device checks before patient handover.

The subsequent full-data export snapshot passed 160 Swift Testing tests plus four XCTest checks. The final English/German UI flow passed, including largest German app text, share-sheet cancellation and a second export. The synthetic ZIP was independently extracted and all six manifest checksums verified, including raw records, a movie, study settings and the archive README. Missing-video reporting, failure cleanup, queued saves, credential exclusion and interrupted-export cleanup were checked; the

simulator contained no leftover ZIP after dismissal. An unsigned generic-iPhone Release build succeeded. A real-phone export/AirDrop, large-volume storage-pressure check and recipient-side verification remain physical checks.

Daily trends now use filtered medians with protocol/version/context isolation. New camera signals use live palm/aspect normalization; historical samples are not rewritten. See Results for eligibility, units and remaining physical validation.

Outstanding decisions include reflection weekdays/reminders, licensed PDQ-8 content, Apple Health integration, validated task-specific composites, formal privacy/retention and consent-based research transfer. Documentation does not approve or implement them. Backend configuration is disabled at launch, but explicit session Finish can enqueue measurements locally; do not enable uploads without reviewing that queue and the full data contract.

Data Formats and Exports

Implementation reviewed 16 October 2026. This describes current files and manual exports, not an approved research-transfer policy or a verified backup service. Source models and serializers are linked below; there is no single global schema version.

Files and Time Formats

All locations are relative to the iPhone app container.

Data	Location	Format and timestamps
Stimulation/app events	`Documents/vcr/logs/events.jsonl`	One JSON object per line; `ts` is ISO-8601 with fractional seconds
Journal	`Documents/vcr/journal/journal_entries.json`	JSON array; dates use Foundation's default numeric seconds since 2001-01-01 UTC, not Unix seconds
Movement sessions	`Documents/vcr/tasks/sessions.jsonl`	One session per line; dates use the JSON encoder's ISO-8601 strategy
Private camera recordings	`Application Support/vcr/tasks/videos/<trial UUID>.mov`	Silent movie linked by trial metadata
Generated movement exports	`Documents/vcr/tasks/exports/`	Separate JSON/CSV files generated on request

Settings, completion counters, calibration and the dormant upload queue also use UserDefaults. They are not all included in measurement exports. Preserve time zones when analysing data: wall-clock recording times identify the event, while local calendar dates group it for the UI. Export filenames are not authoritative recording timestamps.

Events

EventStore.swift writes `ts`, `type`, `tag`, `message` and optional `details` (string-to-string values). Writes are serialized. Events describe the app's actions/state and reported interruptions; they do not independently measure physical motor timing or delivered stimulation dose.

Stimulation Configuration

`vcrStimulationProfile.v1` is one JSON-encoded Data value in UserDefaults, defined by StimulationProfile.swift. Fields are `version` (1), `leftAmplitude`, `rightAmplitude`, `frequency` (cycles/second), and `pattern` (`shuffled` or `simultaneous`). SDK amplitudes are not calibrated physical units. Pulse length remains 100 ms, with four slots per cycle. A missing profile retains previous defaults; an unreadable, invalid or unsupported profile blocks stimulation pending staff review.

The JSONL `stimulation_profile_saved` event includes the new profile and `previous_` values; `stimulation_started` includes the session snapshot, `position`, `effectiveAmplitude` and `durationSeconds`, also on resume/join. `finger_check_started`, `finger_check_cancelled`, `research_unlocked` and `research_locked` record relevant app actions. All parameter detail values are strings. These events do not identify/authenticate an experimenter or prove hardware delivery. A new

glove/resume uses the same snapshot as the active session; locking controls leaves the saved profile unchanged. Existing free-text `app_event` records remain compatible. Measurement JSON/CSV exports do not contain this configuration or the stimulation log; the researcher full-data ZIP does.

Journal

JournalModels.swift and JournalStore.swift define entries with a UUID, event `date`, `type`, optional answers and optional `createdAt`/`updatedAt`. Types are `dailyCheckIn`, `medication`, `symptom`, `note`, `stimulation`, and a reserved `pdq8` case. The reserved case does not mean a PDQ-8 questionnaire is available.

New two-question check-ins have `checkInVersion: 3`, mood and activity impact. Older check-in versions and medication/symptom fields remain readable; an edit does not manufacture missing historical timestamps or silently upgrade their version. Medication entries can include name, dose, event, motor state, factors and a note. Symptom episodes and notes remain separate records.

Journal edits preserve IDs, original creation times and unknown stored JSON fields. Writes are atomic. Unreadable storage or duplicate IDs prevent overwriting the journal; a failed save must leave the user's answers available for retry. Optional/absent answers are not equivalent to "none", "not at all" or zero.

Movement Sessions and Trials

The contract is MovementModels.swift; persistence is TaskSessionStore.swift.

Level	Main fields
Session	`id`, `patientId`, `date`, `stimulationContext`, `trials`
Trial	`id`, `taskType`, `side`, `source`, `stopCondition`, `startedAt`, optional `startUptime`, `samples`, `metrics`, optional `medicationTiming`, `recordedDurationSeconds`, `video`, `measurementVersion`, `fingerTappingAngle`, `fingerTappingAngleMetrics`, `fingerTappingReferenceSegment`
Sample	`t` (seconds relative to trial start), `value` (task/source-specific signal)
Video reference	Relative `fileName`, `byteCount`, optional `timing`; no movie bytes in JSON

Task codes `3.4`, `3.5`, `3.6` identify finger tapping, hand opening/closing and pronation/supination. They are reference identifiers, not assigned MDS-UPDRS ratings. Hands are `left`/`right`; sources are `camera`, `watchMotion`, `synthetic`. Never pool these as interchangeable measurements.

Camera signals are normalized landmark distances: thumb-to-index distance for tapping, and mean fingertip-to-palm-centre distance for opening/closing, divided by hand scale. They are not millimetres. Watch capture uses forearm-axis rotation rate in radians/second; physical orientation still needs validation. Sample times use a monotonic clock rebased to the recording start. Watch rebasing preserves sample spacing but does not establish zero transport delay or precise absolute cross-device synchronization. `recordedDurationSeconds`, when present, is elapsed recording time, distinct from the last detected sample time.

New trials optionally store `measurementVersion`: camera `camera-live-palm-aspect-v2-analysis-v3-local-excursions`, Watch `watch-rotation-rate-v1-analysis-v1`. Earlier live-palm camera recordings retain

`camera-live-palm-aspect-v2-analysis-v1` or `camera-live-palm-aspect-v2-analysis-v2-gap-aware`. Synthetic/unversioned and historical trials may omit it. Camera v2 divides aspect-correct distances by the live frame's wrist-to-middle-MCP length; previous code could use a saved image-space scale. Do not rescale historical values or pool versions. Medication/video copies and JSON export retain the field. Metrics CSV appends `measurement_version` after the original 28 columns; blank means unversioned, not the current method. Raw-sample CSV remains unchanged and can join by trial ID. Calibration remains a positioning/readiness aid, not a physical length measurement.

Context

- New stimulation choices: `preStim`, `postStim`, `noStimPlanned`. Historical `baseline` and `unspecified` remain readable; never relabel them as no stimulation today. These are reported categories, not verified dose/timing.

- Medication status: `takenAt`, `noneToday`, `unsure`; absence means unanswered/legacy. `takenAt` and `confirmedAt` are timestamps. Optional journal ID/name/dose retain their source meaning. Do not infer medication ON/OFF or intake from a reminder.

- Use `Trial.startedAt` for each recording's timestamp, rather than the encompassing session date.

Auxiliary Finger-Tapping Angle

New camera finger-tapping trials optionally include `fingerTappingAngle`, separate from the unchanged distance `samples`, `metrics` and `measurementVersion`. The auxiliary object contains `measurementVersion`: `vision-thumb-wrist-index-angle-aspect-v1`, `unit`: `degrees`, `timeReference`: `seconds_since_trial_start`, and `samples`: objects with `t` and optional `degrees`. Missing/null `degrees` means the processed frame had unavailable landmarks/geometry, not zero movement. Zero degrees is a valid measured aperture. An empty auxiliary array means capture was enabled but no in-window frames were received. A missing auxiliary object means no angle channel was recorded, as on older trials, Watch tasks and hand opening/closing.

The unsigned 2-D thumb-tip/wrist/index-tip angle uses the wrist as vertex and corrects normalized x coordinates by the processed image width/height. It is not a 3-D joint angle or a clinical score. It does not replace the distance signal or feed patient plots, live rhythm, stopping rules or existing metrics. Angles and available distance values from a frame use exactly the same relative timestamp. Angle samples may exist without distance samples when palm-scale landmarks are missing. Entirely dropped/unprocessed frames appear only as time gaps; unavailable processed frames retain an entry. No interpolation is applied.

Medication/video copies, accepted saves, Retake replacement and video deletion retain the auxiliary channel. JSON export includes it; the full-data ZIP retains it inside raw `tasks/sessions.jsonl`. The separate angle CSV includes IDs, task/side, `t_sec`, `angle_degrees`, `angle_valid`, `measurement_version`, `unit` and `time_reference`. `angle_valid` means an angle was calculable, not that task performance, hand identity or clinical interpretability was verified. Missing angles have a blank value and `false`; zero has `0.0` and `true`.

Settings > Data offers the bilingual angle CSV action only when an auxiliary recording exists. Existing raw-sample and metrics CSV columns are unchanged. An enabled take with no received frames exports an angle header with no rows. Historical distances cannot uniquely reconstruct angles; video reprocessing and external healthy-reference comparisons remain separate stages. See the implementation stages and evidence.

Metrics and Missing Data

Results eligibility accepts timed takes with at least 10 seconds of continuous tracking and known elapsed time (0.1-second frame-boundary tolerance); the planned timer need not finish. Matching early-stopped takes contribute to the normal daily median and trend line. Gap-aware camera values can qualify despite missing tracking elsewhere; other methods retain their gap exclusion. This display rule does not change saved metrics, elapsed time, task targets or export fields.

An optional local `movementTrendIdentityLinks` preference stores explicit session-UUID-to-TEAM-ID display mappings with explicit trial-UUID sets for confirmed earlier unassigned tests. No mapping is created automatically. Original session/trial JSON and CSV keep their saved IDs; the research archive includes the mapping as JSON-encoded Data in `study-settings.plist`. Mappings apply only while the original session is unassigned, never to other assigned identities or newly created session UUIDs.

Stored metrics are `cycleCount`, `frequencyHz`, `meanAmplitude`, `amplitudeDecrementSlope`, `rhythmCV`, `pauseCount`, `onsetLatencySec`, `qualityIndex`. The last is an unvalidated heuristic retained for compatibility and hidden from patient results. Rate is displayed as taps/repetitions per second; rhythm variability is interval CV multiplied by 100.

From 19 October 2026, new camera analysis splits samples when consecutive valid timestamps are more than 0.2 seconds apart. This is a continuity tolerance, not a clinical threshold; the separate two-second UI warning remains unchanged. Smoothing and cycle detection restart in each segment with at least four samples. Rate is the count of within-segment cycle intervals divided by their summed time; rhythm and pauses use only those intervals. A visible stationary hand still produces samples, so an observed pause is not removed as tracking loss. With gaps, amplitudes use only event-to-event cycles; decrement pools within-segment mean-normalized slopes, weighted by the number of amplitude comparisons. Neither amplitude nor decrement bridges missing observations. Results require actual within-segment evidence, not just cycles scattered across disconnected fragments. Onset retains its existing first-observation-relative definition; initial unseen movement cannot be inferred. No interpolation, saved-data migration or automatic recalculation of older stored metrics is performed. Watch/synthetic analysis and the auxiliary angle measurement definition are unchanged.

Camera analysis v3 also replaces fixed low/high signal levels with local-extremum directional hysteresis. A rise of 20% of the segment's smoothed 5th-to-95th percentile range records an event after a fall of that excursion. Each segment starts unarmed, so an initial partial rise is not accepted as a cycle. The existing 0.12-second minimum event spacing remains unchanged. The excursion equals the earlier high-minus-low band width, but its anchor follows the local extremum, so baseline drift need not hide a repetition. Smoothing, gap boundaries, raw distances and auxiliary angles are unchanged; amplitude and decrement are not normalized to a moving local range. The former gap-aware detector remains available for v2 prerequisites. Results explicitly project known live-palm camera v1/v2 raw signals through v3 for comparable trends; current v3 results use their saved metrics. This transient projection is not serialized and never changes a trial's stored version, metrics, raw channels or JSON/CSV exports. Unknown or unversioned capture methods, Watch and synthetic sources are not projected. Different or missing study IDs remain separate.

Local excursions are signal events, not verified complete clinical repetitions. Motion below the excursion tolerance, landmark errors, drift/jumps and rapid movement near the sampling limit can still affect counts. A long detected interval is not proof of a physical pause or freezing. See the limited video checks in the validation record.

An empty/short/gapped signal can still be saved. Stored numbers alone do not prove an analysable recording; apply the current display-validity checks and keep unavailable values separate from real zeros. These metrics do not establish normality, clinical severity or treatment effects; see Results.

Saving and Video Deletion

New camera videos on iOS 18.2+ can retain `video.timing`: version 1, method `avcapture-first-buffer-pts-v1`, `firstFrameTrialTimeSeconds` and `firstFrameMoviePTSSeconds`. Map a sample at `t` to movie PTS with `t - firstFrameTrialTimeSeconds + firstFrameMoviePTSSeconds`. Signed offsets are preserved; absent timing is not zero. Older recordings, older iOS callbacks and failed timing inspection retain no inferred anchor. Sessions JSON and full-data snapshots include it; existing CSV schemas stay unchanged. See movie/signal timing for the contract and pending physical verification.

Each completed guided take is saved without requiring all six task/hand combinations. Re-saving the same session updates its line by UUID. Retake replaces the existing task/hand only after the new save succeeds; session files are therefore not an append-only audit trail of every discarded take.

Private movie copy and atomic session write are coordinated. Failed saves retain the temporary draft for retry; measurement-only saving is explicit. A crash between movie copy and metadata commit can leave an orphan movie, so automatic recovery and verified backup are still pending.

Deleting a video keeps the trial, samples and metrics, removes its reference and records a raw JSON `videoDeletedAt` tombstone. This prevents a stale session checkpoint restoring the reference. Copies already shared elsewhere are not removed. Deletion preserves unknown JSON fields and unrelated lines. Ordinary session upserts preserve unrelated lines and deletion tombstones, but do not guarantee retention of every unknown field in the rewritten session. Preserve raw originals for research migrations.

Movies request complete iOS file protection and are excluded from automatic backup. App removal/device loss can lose data. This is file deletion, not a forensic secure-wipe guarantee. See Video storage.

Export Coverage

TaskSessionExporter.swift produces:

Export	Contents	Does not contain
Sessions JSON	Pretty-printed array of decoded sessions, trials, signals, metrics and video references	Video bytes, event log, journal, UserDefaults, unknown raw JSON fields/tombstones
Metrics CSV	One row per trial, IDs/context/timestamps, calculated metrics and medication fields	Raw samples, movie bytes, journal/events
Samples CSV	`session_id`, `trial_id`, `task`, `side`, `t_sec`, `value`	Metrics, medication details, movie bytes, journal/events
Finger-tapping angle CSV	One row per processed angle frame, explicit degrees/validity/version/time reference; joins by session/trial ID	Distance metrics, inferred clinical scores, healthy limits or movie bytes
Video sharing	Explicitly selected private movie files	A complete data manifest or automatic research upload
Researcher full-data ZIP	Raw journal, events/Watch handshake log, sessions JSONL, all retained task movies, allowlisted study settings and file manifest	Unsaved drafts, live recordings, previous generated exports, credentials or automatic upload

The metrics CSV currently uses the last sample time for `duration_sec`, or zero for no samples. It does not use `recordedDurationSeconds`; do not treat that column as elapsed recording time when tracking was interrupted. JSON retains the elapsed field when available.

The trend display can estimate duration for older, unversioned camera duration tests whose elapsed field is absent and whose signal covers approximately the full requested test without a gap warning. This estimate is labelled in Results/history and never fills the missing JSON field or changes CSV values. Explicit short/invalid durations are not overridden; incompatible measurement methods and participant IDs remain separate. See Results for the exact display rules.

The separate movement JSON/CSV exports are generated from decoded records, not a byte-for-byte archival copy of storage. Malformed lines omitted during loading are not recovered into those exports. Verify expected session/trial counts, timestamps and video matches after copying, before deleting any originals.

Researcher Full-Data ZIP

Unlock Study team, open Study setup & logs, then tap Share all recorded data. FullDataExporter.swift prepares one ZIP and opens the system share sheet; the operator chooses the destination. No data is automatically sent or deleted. Export is unavailable during an active, paused or starting stimulation session to avoid competing with stimulation timing.

The ZIP's `vCRGlove-data` folder contains `README.txt`, `journal/journal_entries.json`, `logs/events.jsonl`, optional `logs/handshake.jsonl`, `tasks/sessions.jsonl`, `tasks/videos/`, `study-settings.plist` and `manifest.json`. Raw storage is copied without model decoding/re-encoding, preserving unknown fields, deletion tombstones and malformed historical lines. All retained UUID-named task movies are included, including orphaned files; bundled demonstrations and unaccepted temporary takes are not.

The settings property list includes patient/study metadata, the encoded stimulation profile, hand calibration, session completion preferences, app/reminder preferences and dormant pending upload data. Data-valued preferences appear as base64 in the XML plist; their original encoded contents are preserved. It is an explicit allowlist, not a dump of all UserDefaults or credentials. Missing settings are not fabricated.

Manifest format 1 records the UTC export interval, original time zone, app version, relative file paths, byte counts and SHA-256 checksums, excluding the manifest itself. Absent stores are listed rather than invented. Missing referenced movies are listed separately and prompt a warning before sharing. An unreadable file aborts preparation; originals remain unchanged. Store snapshots wait behind queued writes; they are sequential, not a globally atomic transaction. Unsaved or concurrently started recordings are not promised in that snapshot.

Video copying and ZIP/checksum work run off the main thread. Temporary share copies request complete file protection and backup exclusion, are removed after share dismissal/failure, and interrupted-export copies are cleared on next launch. Originals remain in their existing stores. File protection is not ZIP password encryption: the shared ZIP contains sensitive data and requires an approved destination. Sharing completion does not prove recipient receipt or a verified backup; backup confirmation, restore/recovery and consent-based transfer remain separate work.

Offline Angle Comparison

In-App Angle Summary

New camera tapping trials also save optional `fingerTappingAngleMetrics` with `calculationVersion` (`vision-angle-timestamp-gap-aware-v1`), `intervalCount`, `openingPeakCount`, `trackedSeconds`, and optional `meanOpeningAngleDegrees`, `meanAbsoluteAngularSpeedDegreesPerSecond`, `meanRateHz`, `periodCV`. Absent metric values mean unavailable, not zero. See the native calculation contract. JSON/JSONL and full-data exports preserve the field; existing CSV schemas and distance metrics are unchanged. Historical angle summaries are display-only, not a stored migration.

First-Ten-Tap Comparison Window

New tapping trials can additionally save `fingerTappingReferenceSegment`: `selectionVersion` (`vision-first-ten-consecutive-complete-taps-v1`), `sourceAngleVersion`, `detectorVersion`, `completedTapCount` (10), `firstSampleIndex`, `endSampleIndexExclusive` (zero-based half-open indices into the full angle array), `startTrialTimeSeconds`, `endTrialTimeSeconds`, and `openingPeakTimes` (eleven trial-relative seconds). See the selection contract. It identifies a comparison segment without trimming the saved 30-second trial. Absent means unavailable/not recorded, not zero taps or a failed overall task. Medication/video copies, JSON/JSONL and full-data exports retain it; existing CSV schemas are unchanged. No legacy migration or healthy band is implied.

Offline Research Tool

Research-only angle comparison is available through the offline review tool. It reads exported sessions without writing app records, preserves stored metrics alongside current replay results, and keeps reports outside Git. It does not infer missing legacy angles or provide healthy ranges. Associated video file/byte checks are not independent accuracy or synchronization validation.

Upload Boundary

There is no configured backend at app launch. Explicit session Finish still calls SessionUploader.swift; without configuration it queues encoded measurements locally in UserDefaults (`vcr.uploader.pending`). Per-take automatic checkpoints do not request upload. This prototype transfers neither video bytes nor journal/event files, and the bundled server does not yet preserve the complete medication/video contract.

Do not enable or flush that queue until endpoint security, consent, identity, deduplication and field completeness are reviewed. A successful HTTP response alone is not a verified backup. See backend reference.

Release Checklist

Use this for each deliberate TestFlight release. It documents the process; it does not authorize an upload or patient deployment.

Version and Build

Version names a milestone; build identifies an uploaded binary. For example, `0.5.0 (2)` can contain fixes after `0.5.0 (1)` without changing the milestone. Setting a version does not freeze development.

The current project snapshot has phone and Watch version `0.5.0`, build `1`. This is not a record of the latest App Store Connect upload. Check the actual uploaded build before choosing the next number. Keep shipped phone/Watch version and build aligned in Debug and Release; test bundles have separate version settings.

Use a new, increasing build for each upload; do not reuse a previously uploaded number for that version. Apple associates uploads using bundle ID, version and build. Record the final uploaded values if Xcode manages/increments them during distribution. See Apple's upload guidance.

Before Archiving

[ ] Agree on the release scope and tester group. No unapproved clinical/privacy changes.

[ ] Review `git status --short` and the diff. Include intended source, bundled assets, tests and documentation; exclude raw source videos, patient exports, credentials and personal workspace settings.

[ ] Update version/build and release notes. Confirm ICNS signing, bundle IDs and embedded Watch identity.

[ ] Run relevant unit/UI tests using the development guide. Record their exact commit and results; do not carry forward old passes as new verification.

[ ] Perform the applicable physical smoke tests, including English/German, largest app text, early Stop, saving/relaunch, playback/export and deletion cancellation. Mark Watch/glove tests deferred if not actually performed.

[ ] Verify both hands' saved stimulation amplitudes and approved pattern/frequency in person before handover. Research unlock must not create another tab, and patient Start/reconnect must not restore the old preset. Check startup cancellation and hardware pulse timing; software tests alone do not validate delivery.

[ ] Verify update compatibility on disposable data. Preserve IDs, old journal fields, recordings and deleted-video markers. Do not delete a study app to resolve an install error without a verified data copy.

[ ] Review known limitations and privacy/consent requirements before adding patients. A successful build/upload is not study deployment approval.

Commit and Tag

The order to remember is **inspect, stage, inspect staged, commit, inspect, push**. Run from the repository root. Select paths deliberately instead of using `git add .` on a mixed working tree.

```
git status --short
git diff --stat
git diff
git add path/to/changed-file
git diff --cached --stat
git diff --cached
git commit -m "Describe the tested change"
git log --oneline -3
git status --short
git push
```

Replace `path/to/changed-file` with the intended files before running the staging command. Unrelated untracked files may remain, but must not be included accidentally. A release archive should correspond to a committed snapshot without uncommitted tracked app changes.

After verifying the release commit, create an annotated tag, for example `v0.5.0-build.2`. The tag identifies the source used, not proof of clinical validation. Do not move a published tag to another commit.

```
git tag -a v0.5.0-build.2 -m "TestFlight 0.5.0 build 2"
git push origin v0.5.0-build.2
git rev-parse HEAD
```

Use the actual version/build, check that the tag is new, and push only the intended tag. If distribution changes the final build number, reconcile the release record/tag before sharing the build.

Archive and Distribute

1. In Xcode select the `vCRGlove` scheme and a generic physical iOS destination, then Product > Archive. Review the archive's version/build and embedded Watch app.
2. In Organizer validate the archive and review warnings. A missing vendor dSYM affects crash symbolication; obtain symbols matching the framework UUID from the vendor rather than inventing an empty dSYM. Keep the archive and available symbols in approved storage.
3. For internal/external TestFlight flexibility choose **App Store Connect** (called **TestFlight & App Store** in some Xcode flows). **TestFlight Internal Only** restricts that upload to internal testers. Uploading does not publish the app publicly. See Apple's distribution tutorial.
4. After processing, verify the actual version/build in App Store Connect. Review encryption declarations against all bundled dependencies; the current project sets `ITSAppUsesNonExemptEncryption = NO`, which is not a claim that the OS uses no encryption.
5. Add the build to the agreed TestFlight group with concise "What to Test" notes and known limitations. Do not give patients developer/admin roles merely to distribute a build. Internal testers require appropriate App Store Connect access; external testing is a separate workflow. See Apple's internal testing guide.
6. Install the distributed build and perform a short smoke check. Record the uploaded binary's identity and distribution result, not just the local Xcode run.

Rollback Without Data Loss

Stop broader rollout when a regression is discovered and keep the failing build/logs for diagnosis. Do not uninstall a study app or clear its container as a rollback step.

Prefer a reviewed fix or `git revert` on a new branch/commit, then release with a new build number. Check persisted-data compatibility: reversing code does not reverse a storage migration. Test recovery

against a copy before touching patient data. Avoid `git reset --hard`, force-pushing shared history or moving published release tags.

An older TestFlight build is an option only if still available and compatible with the current stored data. Verify that on disposable data first; never assume it restores lost records or videos.

Release Record

Keep one completed record per distributed build alongside team release notes. Store device logs/screenshots and archives in approved institutional storage, not patient data in Git.

```
Version/build:
Source commit/tag:
Release date and responsible reviewer:
Xcode/macOS and SDK:
Phone/Watch models and OS tested:
Automated tests (command/result location/commit):
Physical checks (pass/fail/deferred and evidence):
Changes to test:
Known limitations and deferred checks:
Data compatibility/export checks:
Archive/symbols location:
App Store Connect processing result and actual uploaded build:
Tester group and distribution date:
Follow-up issues/backlog IDs:
```

vCRGlove Testing Checklist

Use this checklist when testing the vCRGlove app on study iPhones with bHaptics TactGlove DK2 gloves. Record the phone model, iOS version, app commit/tag, glove IDs, glove battery levels, date/time, and tester initials for every test session.

Reviewed 15 October 2026. Use disposable non-patient data and approved bench/study procedures. Record each check as pass, fail or deferred with evidence; this checklist does not mean the checks have been performed. See the development guide for simulator coverage and the release checklist for distribution.

Test Session Metadata

- Tester:
- Date:
- iPhone model:
- iOS version:
- App branch/tag/commit:
- Installed version/build and installation source (Xcode/TestFlight):
- Xcode/macOS (if testing locally):
- Watch model/watchOS and companion version, if tested:
- Left glove ID:
- Right glove ID:
- Starting left glove battery:
- Starting right glove battery:
- Notes:

1. App Launch and Navigation

1. Launch the app from a fresh closed state.
2. Confirm the patient tab order is vCR, Journal, Movement, Settings.
3. Open Settings.
4. Unlock Research Mode with the current research password.
5. Confirm there are still exactly four tabs, in the same order.
6. Open the existing vCR tab and expand Stimulation parameters.
7. Lock researcher controls in Settings and confirm the parameter panel disappears. Relaunch: editing stays locked, saved parameters remain.

Expected result: one vCR tab for patients and staff. The discreet Study team unlock entry is below Contact/Help. No old visibility preference can restore another tab.

2. Patient Glove Scanning and Pairing

1. Turn both gloves off.
2. Open the patient vCR tab.
3. Press Scan for Gloves.
4. Confirm both glove cards remain gray/disconnected.
5. Turn on only the left glove.
6. Press Scan for Gloves again.
7. Confirm only the left glove becomes ready.
8. Turn off the left glove and turn on only the right glove.
9. Press Scan for Gloves again.
10. Confirm only the right glove becomes ready.
11. Turn on both gloves.
12. Press Scan for Gloves.
13. Confirm both gloves become ready.

Expected result: no stale or nonexistent glove should appear as ready. Left and right status must never be mixed.

3. Patient Test Buzz Behavior

1. With one glove ready and no stimulation running, tap the glove icon.
2. Confirm the corresponding physical glove buzzes briefly.
3. Start a vCR session.
4. Tap the glove icon during stimulation.
5. Confirm no extra test buzz occurs during stimulation.

Expected result: test buzz uses that hand's saved amplitude and works only when the glove is ready and no session (including paused/startup) is in progress.

4. Patient vCR Start, Pause, Resume, and Stop

1. Connect at least one glove.
2. Press Start vCR.
3. Confirm stimulation starts on all ready gloves.
4. Confirm the timer, progress bar, and running status appear.
5. Press Pause.
6. Confirm stimulation pauses on all active gloves.
7. Press Resume.

8. Confirm stimulation resumes.

9. Hold the Stop control until it completes.

10. Confirm stimulation stops and the UI returns to ready state.

Expected result: start, pause, resume, and hold-to-stop are clear, responsive, and do not crash the app.

5. Adding a Second Glove During Stimulation

1. Turn on and connect only one glove.

2. Start vCR stimulation.

3. While stimulation continues, turn on the second glove.

4. Press Scan for Gloves if needed.

5. Confirm the second glove connects.

6. Confirm the second glove automatically joins stimulation with the remaining time.

7. Repeat while the session is paused:

- Start with one glove.

- Pause the session.

- Connect the second glove.

- Confirm the second glove joins in paused state.

- Resume and confirm both gloves stimulate.

Expected result: adding a second glove must not interrupt the first glove.

6. Glove Disconnection During Stimulation

1. Start stimulation with both gloves.

2. Turn off or disconnect one glove.

3. Confirm stimulation continues on the remaining glove.

4. Turn the disconnected glove back on.

5. Confirm it can rejoin stimulation.

6. Repeat the test but disconnect both gloves.

7. Confirm the session is marked interrupted because no active gloves remain.

Expected result: one lost glove should not end the whole session; zero active gloves should interrupt the session and log the reason.

7. Full 4-Hour vCR Session Completion

1. Start with both gloves charged sufficiently.

2. Start the default 4-hour vCR session.

3. Keep the app open for the full session.

4. Avoid touching the phone except to confirm the app remains visible.

5. At session end, confirm stimulation stops automatically.

6. Confirm the patient sees a completion message.

7. Confirm the calendar/day status updates as expected.

8. Export or inspect logs and confirm the session start and completion were recorded.

Expected result: full 4-hour stimulation completes without stutter, crash, premature stop, or missing completion message.

8. Two-Session Daily Use Scenario

The 2 x 2 hour plan is available in Settings. Record the selected plan and use the approved test procedure; do not change clinical session durations for convenience.

1. Select 2 x 2 hours and run the first session.

2. Confirm the app does not show full-day completion after only the first 2-hour session.

3. Run the second 2-hour session.

4. Confirm completion appears only after both sessions are finished.

Expected result: daily completion should reflect the configured daily stimulation plan.

9. Background and Foreground Timing Compromise

1. Start stimulation with at least one glove.

2. Press the Home button or swipe to leave the app briefly.

3. Return after 2-5 seconds.

4. Confirm the app logs that timing may have been compromised.

5. Repeat with longer background durations:

- 10 seconds

- 30 seconds

- 2 minutes

6. Try these background scenarios:

- Open another app.

- Lock the screen.

- Receive a notification.

- Open Control Center.

- Answer or dismiss a call if safe to test.

7. Record whether stimulation continues, stutters, pauses, or stops.

Expected result: inactive transitions are logged; entering background also shows a timing warning and requests a finite grace period. Neither guarantees uninterrupted hardware timing. Document actual glove behaviour, warning text and timestamps separately for each scenario.

10. Stability and Crash Stress Testing

1. Rapidly switch between vCR, Journal, Movement and Settings tabs during no stimulation.
2. Repeat during active stimulation.
3. Rapidly start and stop scanning.
4. Connect and disconnect gloves repeatedly.
5. Unlock/lock researcher controls in Settings; the tab count and active stimulation must not change.
6. Open Journal while stimulation is running.
7. Add a daily check-in while stimulation is running.
8. Add medication, symptom, and note entries while stimulation is running.
9. Leave the app idle for at least 30 minutes with no stimulation.
10. Leave the app idle for at least 30 minutes during stimulation.

Expected result: no crashes, no UI freezes, no unexpected stimulation stop unless explicitly caused by disconnection/background constraints.

11. Research Mode Testing

1. Unlock Research Mode from Settings.
2. Open the existing vCR tab and expand Stimulation parameters. No log should appear on this page.
3. With the approved bench procedure, set distinct left/right amplitudes, cycle frequency and pattern; press Save. Merely editing must not affect saved values.
4. Lock controls and relaunch. Confirm editing starts locked. Unlock and verify saved values, without an extra tab.
5. Start with real gloves. Check each hand's startup check, idle buzz and active amplitude against the intended configuration and motor mapping. Do not infer delivery from the displayed percentage.
6. Confirm controls cannot save during startup, running or paused states. Locking controls must not alter stimulation.
7. Pause/resume, disconnect/reconnect one glove and add a second glove; all must retain the original session profile and remaining duration.
8. Hold Stop during the finger check. Confirm every motor stops and stimulation cannot restart a few seconds later. Disconnect a glove during that check and confirm a fresh Start is required.
9. Repeat Stop/Pause at different points in a cycle; no queued pulse may fire after stopping/pausing. Check timing on both gloves, including the 2.5 Hz boundary with 100 ms pulses.
10. Test both patterns under the approved protocol: shuffled motors 0-3 once per cycle; legacy simultaneous array on each of four slots per cycle. Verify physical finger mapping and delivered pulse durations with suitable hardware measurement.

11. Open Study setup & logs in Settings. Inspect/share the event log: saved profile changes include old/new values; start/resume/join records include effective per-hand amplitude, frequency, pattern and duration. The visible log must not flood with repetitive raw BLE snapshots.
12. Test invalid/corrupt configuration only in isolated fixtures: delivery must remain blocked until staff save a valid profile. Recheck 4-hour and 2+2-hour completion separately.

Expected result: saved staff configuration is consistently applied, cannot change mid-session, and is not reset by patient Start. Simulations are not evidence of clinical safety, hardware timing or secure authentication; the existing password remains a basic UI gate.

12. Journal and Daily Data

1. Open Journal.
2. Select today in the calendar.
3. Complete the mood/activity-impact check-in. Confirm answers are in severity order and saving shows today's check-in status.
4. Edit the check-in, then use its Medication/Log symptoms shortcuts. Filled answers must save before navigation; an empty shortcut must not create a check-in.
5. Add and edit a medication event. Non-usual doses reveal an optional note; changing the answer must not silently discard typed notes or imply a dose was taken.
6. Add and edit a symptom episode and a note using the direct Journal dashboard links.
7. Start and stop or complete a vCR session.
8. Return to Journal and confirm all entry types appear for the day.
9. Restart the app and confirm entries/edits persist without duplicates. Check a past day and future-date restrictions. Repeat in English/German and at the largest app text size; inspect symptom choices and all save buttons.
10. Copy the app's Documents data via an approved method and inspect `vcr/logs/events.jsonl` and `vcr/journal/journal_entries.json`. Preserve originals. Journal dates use Foundation's numeric reference-date encoding; see Data formats.

Expected result: entered data and edits persist, shortcuts do not lose answers, and old entries retain their meaning. Reflection schedules, medication reminders and PDQ-8 are pending features, not acceptance requirements for the current build.

13. Battery and Low-Battery Behavior

1. Connect both gloves with high battery.
2. Confirm battery/ready status appears correctly if available.
3. Test with a low-battery glove when available.
4. Confirm low battery is visibly communicated.
5. Confirm stimulation is not started accidentally on a disconnected or empty-battery glove.

Expected result: low battery should be clear to the user and should not be confused with disconnection.

14. Data Review After Testing

After each test session:

1. Copy app files using the approved workflow. Share private videos separately from Settings > Data; movement JSON/CSV does not contain movie bytes.
2. Save logs together with the test metadata.
3. Note any timing compromise warnings.
4. Note any Bluetooth connection failures.
5. Note any crashes, including approximate time and preceding action.
6. Note whether the physical glove behavior matched the app UI.
7. Verify copied session/trial counts, recording times and video linkage before deleting any original. Separate movement exports omit journal/events and movie bytes; the researcher ZIP includes all retained app study data but does not verify recipient receipt or backup. See Data formats.

Full-device export from researcher settings

1. Unlock Study team and open Study setup & logs. Check Share all recorded data is visible in English/German, including large text, and absent from locked patient settings.
2. Export a known non-patient fixture containing journal, stimulation logs, movement samples, saved movies and study settings. The share sheet must offer one ZIP. Cancel once, then export again; no originals may change.
3. Extract the ZIP on the receiving computer. Verify all file sizes and SHA-256 values against `manifest.json`, movie playback, journal dates (seconds since 2001) and trial ISO-8601 timestamps. Check raw legacy fields and deletion markers remain intact.
4. Check empty data stores and old references to missing movies. Missing movies must produce a warning and a manifest list, not silently claim a complete video set. Never include passwords or bundled demonstration clips.
5. Confirm preparation is unavailable during starting, active and paused stimulation. Test a larger set of videos for storage-pressure/error handling and an unlocked physical phone for protected-file sharing.
6. Dismiss the share sheet and confirm temporary export copies are removed. Interrupt preparation/relaunch and confirm cleanup; saved original movies and measurements must remain. Receipt/backup validation is separate from the share-sheet action.

Recommended issue note format:

```
Date/time:
Tester:
Phone/iOS:
App commit/tag:
Glove side and ID:
Steps to reproduce:
Expected behavior:
Observed behavior:
Logs/screenshots:
Severity:
```

15. Movement Recording and Results Smoke Test

1. Calibrate on the physical iPhone and confirm the hand-size result and concise positioning feedback. A calibration suggestion must not block recording.
2. Record finger tapping and hand opening/closing for each hand. Confirm task/side titles, correct silent demo, countdown, camera framing, accessible Stop/Skip controls and live rhythm. Demo playback must start with recording.
3. Stop immediately, then after a few seconds, and complete a full timed recording. Each case must leave the recording state and offer clear next actions, even with no analysable signal.
4. Briefly move the hand out of frame. Check guidance, signal gaps and the continuing 30-second clock. Missing data must not appear as fabricated zero movement. In a new gap-aware camera recording, confirm lost tracking is not counted as an observed pause/rhythm interval; also hold the hand still while visible and confirm genuinely observed pauses remain measurable. Repeat for tapping and opening/closing. New-build physical verification remains required.
5. Open a saved recording, return, change tabs and relaunch. Saved trials/movies must remain available without finishing every task in the set. An unfinished draft and navigation restoration are separate behaviours.
6. Retake one task/hand. The old saved take must remain until its replacement saves successfully; unrelated takes remain unchanged.
7. Check per-trial timestamps and the chosen stimulation/medication context. Unknown context must remain unknown. Confirm medication is not repeatedly asked on every task page.
8. In Results, switch task/hand, rate/size/rhythm and filters. On two or more comparable days, check daily-median lines separated by context. Repeat a test on one day: count/range must update, without creating extra longitudinal observations. Tap the day and open its recordings. Check the metric info button, More details and paginated history. No clinical score, normal range or treatment-effect claim is implied.
9. Verify exported measurement data against saved recordings. Review `duration_sec` versus elapsed recording duration when signal was missing, as documented in Data formats.
10. Test Watch pronation/supination only on a supported physical pair. If unavailable, mark it deferred and verify that the error and Skip action remain reachable; do not infer Watch success from camera-task tests.

Expected result: completed takes survive navigation/relaunch, Stop never leaves the user stuck, and unavailable analysis does not discard a recording. Record hardware accuracy/timing observations separately from software checks.

Cycle-detector v3 physical check (pending): record steady repetitions, smaller repetitions, a visible hold, baseline/hand-position changes and a tracking gap in each camera task/hand. Manually count short unambiguous video segments and compare with exported-signal events using the matching analysis version; allow for video/signal alignment uncertainty at segment edges. Visible partial movements are not automatically complete protocol repetitions. Check that a static hold does not create extra cycles and that smaller/decreasing excursions are not silently interpreted as improvement. Do not infer clinical accuracy from the limited development recordings or present pause flags as confirmed freezing.

New Trend and Camera Checks (Physical Validation Pending)

- Record the same movement at two comfortable camera distances, keeping the same hand plane. Review raw normalized signal and size against video; the synthetic geometry test is not physical accuracy validation. Repeat with each hand, changed lighting and the normal vertical pose.

- Recalibrate and repeat. New size should not depend on the saved image-space calibration scale. Check camera angle, visible-finger changes and missing/occluded joints for artifacts; do not claim perspective invariance.

- Compare full, early-stop, tracking-gap and older unknown-duration records. Older unversioned camera duration tests with absent elapsed time may join a historical line only when finite ordered samples cover the full requested duration (end and span within the documented 0.5-second tolerance), without a gap warning. Confirm the estimated-duration label in the chart/day summary and history. Short/late-start/gapped signals and explicit short/invalid durations must stay separate; no record/export is rewritten.

- Review right-hand finger tapping across the older October test dates: when all dates/timings are selected, the comparable before-stimulation and after-stimulation historical segments should connect. The newer measurement method and newly assigned study ID must remain separate. This physical-device/export review remains pending even after software regression tests pass.

- With no line, verify the explanation matches the data: only one day, different study IDs/methods/protocols, unmatched stimulation timings, or incomplete capture. Check English/German and the largest supported text size; do not infer a motor-performance failure from a missing line.

- Check historical rate/rhythm lines separately from new methods; historical size remains in history with an older-scale label. Do not expect old and new values to join.

- Apply 7/30/90-day and reported medication-hour filters before aggregation. Verify selected-day counts against original trials, not session dates. Unknown medication/context must never mean no medication/stimulation.

- Review English/German layouts at standard and largest text sizes, VoiceOver day selection, legends and reachable history/filter controls. Test after midnight/time-zone changes and on a supported physical Watch pair when available.

Auxiliary Finger-Tapping Angle (Physical Checks Pending)

- Record both hands through the guided flow and the standalone task route when available. Verify JSON contains a separately versioned `fingerTappingAngle` in degrees, without changing distance samples/metrics or the patient plots.

- Briefly hide the hand, return and close thumb/index. Check unavailable angle entries remain missing, while a valid zero angle is not discarded. Processed missing frames and unprocessed time gaps are distinct; neither pauses the clock.

- Stop immediately, stop after a few seconds, finish 30 seconds and Retake. Confirm saved trials retain the right angle channel on reopen, including when their movie is deleted; a failed replacement must not lose the prior take.

- Export full-session JSON and angle CSV in Settings > Data, then the researcher full-data ZIP. Compare IDs/timestamps, degree units, missing entries and raw JSONL. Old metrics/sample CSV formats must remain unchanged. With only old recordings, the angle CSV action must not appear.

- Check upright/front-camera geometry at different comfortable distances and hand planes. Mirroring invariance does not identify the correct anatomical hand; inspect side labels against the actual task. Perspective/tracking and normative accuracy are not established by simulator tests.

16. Private Videos and Sharing Smoke Test

1. Record on the physical camera; verify silent, upright videos and correct task/hand linkage in Settings > Data.
2. Open a video, return and share explicitly to an approved test destination. Confirm no automatic Photos copy or Photos permission prompt appears.
3. Select multiple videos using long press/Select, then test Share and cancel Delete. Cancellation must leave all selected files unchanged.
4. Confirm deletion using disposable clips only. The selected videos disappear while their movement samples, metrics and context remain. Other videos are unaffected.
5. Relaunch and confirm a deleted reference does not reappear. A previously shared copy remains outside the app's deletion control.
6. Check low-storage/write failures and locked/background behaviour using a controlled test setup. Do not delete live data or fill a study phone to force failure.

Expected result: sharing/deletion is explicit, failure notices preserve retry options, and measurement retention is independent of video retention. Physical file protection, long-term storage, recovery and verified backup require separate evidence.

17. Study Information and App Privacy

1. Confirm the primary Settings order: Profile, General, Instructions, vCR Settings, Reminders, Data, Information about the study. General contains language and text size; neither the old Profile text-size selector nor the globe shortcut remains. Change every text size and language, then relaunch and verify the choices are retained. The old Privacy & Data entry must be absent. Visit each Info topic; Contact must be last. There must be no consent switch, permission prompt or upload action.
2. Repeat in English/German and at the largest app text size. Switching language in General must keep that page and the Settings tab open, translate the page immediately and preserve Back navigation. Profile must show the study ID without avatar placeholders or icon-selection wording. Read the full privacy notice through its version footer; text must wrap and remain scrollable above the tab bar.
3. Compare the retained German privacy text with the current Word source identified in Study information. Only the explicitly approved lab-video paragraph is omitted; all rights, exceptions, legal contacts and definitions remain.
4. Open Contact directly. Confirm existing phone/email links and the W36 institutional address, without a troubleshooting picker, conditional help introduction or Still Need Help? heading. Personal names must not appear on general study/contact pages; legal privacy contacts retain their names. Verify the separate troubleshooting route still works.
5. Open Devices and app and verify that all relocated data/video text matches actual local capture, backup exclusion, sharing and deletion behaviour. Reading the page must not change recording preferences or count as consent.

6. Obtain institutional review of the adapted notice and English legal translation before patient deployment. App-video consent coverage, opt-out and withdrawal handling remain separate pending work.

Expected result: concise study guidance and complete retained legal wording are available offline in both languages. The Info section does not replace study consent or implement future privacy controls.

Consistent Historical Trends and Study Identity (19 October 2026)

[ ] Patient chart has no stacked reanalysis, estimated-duration, missing-value count or daily-median footnotes. Metric info retains explanations; the colour legend, separation link and brief clinical disclaimer remain accessible.

[ ] With a TEAM ID and research unlock, link your own earlier unassigned test recordings, then reopen Trends. Applicable historical lines must appear. Cancel must not change display links. Remove links to reverse the display change. Future unassigned and already assigned recordings must not be relabelled.

[ ] Export JSON/CSV before and after linking: original IDs/data must match. The full research archive must retain the display mapping in its settings snapshot.

[ ] On iPhone, compare known live-palm camera v1/v2/v3 recordings: complete recordings with the same ID, hand, task, context and protocol share a trend.

[ ] Stop a timed camera task after 10 seconds of good tracking. It must use the normal point colour and contribute to the same daily median/line as a completed take. Repeat for tapping and hand opening/closing on both hands.

[ ] Lose tracking, then provide at least 10 continuous tracked seconds: gap-aware camera results can still contribute. A five-second take, isolated samples or disconnected fragments with no ten-second stretch must not qualify.

[ ] Confirm original session JSON/CSV metrics, versions and samples stay unchanged after opening Results; updated display analysis is labelled.

[ ] Unknown methods, different/missing IDs and unusable recordings remain separate. Open the explanation sheet while other trend lines are visible.

[ ] In researcher settings, typing an ID without Save does not change Profile. Cancel an assigned-ID change, then confirm it; check persistence after relaunch. Historical recordings must keep their original ID. An assigned ID cannot clear.

Software checks are recorded in the angle audit. These physical-device checks remain pending; they do not establish clinical accuracy.

Offline Angle Comparison (19 October 2026)

Implemented tooling: see review commands and definitions. Ten synthetic regression tests passed, including actual Swift analyzer replay. The six-trial export review checked paired signals and associated video files, not independent video landmark accuracy. App behaviour is unchanged.

[ ] Independently annotate tap/cycle events in representative left/right videos, including the final two trials with unmatched events. Establish video/signal timing before comparing annotations; do not assume zero offset.

[ ] Verify aperture under hand rotation, camera-distance changes and tracking loss against independent landmarks/annotations, not the shared distance signal.

[ ] Check repeatability across phones, lighting, hands and relevant movement impairment before selecting a smoothing/velocity endpoint.

[ ] Complete external healthy-cohort protocol, population and permission review before proposing any reference band. No patient norms are currently implemented.

Stage 5 desk assessment is documented in Healthy references. The pinned table audit and five synthetic tests passed. Candidate adoption still requires control-identity clarification, protocol/estimator bridging, independent accuracy checks and intended-use approval; the unchecked gate above remains open.

ROC-HCI Feature Subset (19 October 2026)

The offline feature subset leaves app capture, gap-aware analysis and plots unchanged. Thirty offline tests passed. Selected kernel parity passed on 15 synthetic cases and five supported real trials; a sixth stress-test trial was rejected for out-of-range cubic interpolation.

[ ] Review interpolated/selected frame ranges against independent video annotations. Do not treat author gap filling as verified observed motion.

[ ] Match recording instructions and feature definitions to the intended healthy cohort; whole 30 s takes are not ten-tap protocol equivalents.

[ ] Verify the new camera-study and hand-opening/closing benchmarks before adding any patient bands. No camera or clinical accuracy is established here.

Independent Video Review (21 October 2026)

[x] Decode both supplied tapping clips and inspect video-only contact sheets. Complete decoding found 902 frames each; angle streams contain 898/899 entries.

[x] Inspect the same 16-18 s interval at full video frame rate before opening its sample/peak arrays. Eight visible closing phases per hand versus eight/three author-derived opening peaks in the nominal signal window. Left edge-on contact is ambiguous; this is not a synchronized clinical repetition-count comparison.

[x] Preserve originals and keep private frames/annotations outside Git.

[x] Implement an optional first-buffer movie/signal timing anchor for new camera recordings on iOS 18.2+, retaining legacy/failed-inspection fallback.

[x] Initial phone-data check: new right-hand tapping and opening/closing movies include anchors, matching JSON/CSV and visible phase agreement in early/late windows. Older clips without timing cannot verify the new implementation.

[ ] Independently annotate a protocol-matched palm-facing recording and repeat across views. No healthy band or angle-accuracy claim is approved by this check.

See review findings and tools. Eight focused review-tool tests cover pixel geometry, missing/zero values, annotation integrity, event intervals and input manifests. App unit/UI suites were not rerun because this stage changes only offline review tools and documentation.

Movie Timing Metadata (21 October 2026)

The timing contract leaves task timers, movement analysis, stimulation and CSV schemas unchanged. Check source timestamps, not callback arrival time or inferred frame indices.

Software verification passed: 234 unit test methods (302 executions), zero failures/skips, including signed/invalid offsets, legacy/JSON compatibility, metadata retention in an accepted movie/full-data snapshot and timing-inspection fallback. All 38 offline tool tests also passed. These checks do not exercise the real camera's first-buffer callback.

[x] Inspect new tapping and opening/closing exports with `video.timing` and map visible events using the documented formula. Initial 21 October right-hand review: 26 annotated phases matched within visual windows or about one frame. Both movies decoded fully and JSON/CSV fields agreed; see the timing contract.

[ ] Check early Stop, retake, leaving/relaunch and the researcher's full-data export. Timing must follow the correct accepted movie, not a discarded draft.

[ ] Check an older recording and iOS 17-18.1 where available. Missing timing must not prevent saving or be interpreted as zero offset.

[ ] Confirm frame alignment across the clip before using it for angle accuracy measurements. Software tests do not replace a camera-device check.

First-Ten Window And Video Folder (21 October 2026)

[x] Native regression: 243 methods / 311 executions, no failures/skips. First-ten windows preserve full signals and metrics; gaps/interrupted cycles cannot be stitched or restarted later; copies, legacy decode and JSONL retain the optional metadata correctly.

[x] Current saved native summary reproduces exactly in read-only replay of the latest supplied tapping recording. Source exports are not rewritten.

[x] Three bilingual simulator UI regressions passed: the Data page links to one Recorded videos folder; playback, multi-select sharing/deletion, relaunch and angle-export visibility remain functional.

[x] Offline research tooling: 51 tests passed, including display-orientation handling, timestamp pairing and author-kernel comparisons. No patient bands.

[ ] On a phone, save a new full tapping trial and export JSON after relaunch: confirm full 30-second signals plus ten-cycle indices/eleven peak times.

[ ] Repeat with early Stop, tracking loss and another hand. Unavailable comparison window must never block saving or remove ordinary results.

[ ] Verify the Recorded videos folder count, empty state, long-press selection, sharing and deletion on a physical phone. Grouping does not reduce file sizes.

[ ] Independently review first-ten detections and detector/encoding/protocol differences before any healthy-reference feedback. See reference bridge.

Native Tapping Angle Summary (Earlier 21 October Checks)

See calculation and compatibility. The new angle section is separate from the original distance metrics and daily trends.

[x] Unit suite: 239 methods / 307 executions, no failures. Known angles/rates, timestamp scaling, gaps, flat signals, optional fields and legacy decoding.

[x] Simulator UI: angle section in English/German at the largest app text size; row access verified after removing an overriding section identifier. Existing bilingual video save/playback/share/delete/relaunch regression passed.

[ ] On the phone, record both hands; confirm opening angle, angular speed, rate and rhythm appear with units and do not replace existing metrics.

[ ] Try a flat hand, lost tracking and early Stop. Missing values must not become fabricated zeros or cycles across gaps.

[ ] Accept/retake, relaunch and export JSON/full data. Confirm the correct trial retains its versioned angle summary and original samples.

[ ] Compare extracted peaks with timestamp-aligned video; no healthy bands should appear. Clinical and external-control validation remain pending.